

COMP 4107 Midterm 2

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I. SOS GAME

Create an application that plays SOS game.

- Your application will have two activities.
 - The first activity will have the board, one checkbox, and three buttons.
 - * The board will be drawn in width $\times$ height size. The size of a square depends on the size of the screen of the phone.
 - * The checkbox has an "S" on it.
 - * The buttons have an "Show Options", "Load Game", "Restart Game" on them.
 - The second activity will have two spinners, two textviews, two editboxes, and two buttons.
 - * The Textviews have "Width" and "Height" on them.
 - * The spinners have a maximum value of 10 and a minimum value of 3.
 - * The editboxes should be ineditable and will show the value of the spinner.
 - * The buttons have "Restart Game", "Go back" on them.
- The game proceeds as follows:
 - The user starts the game.
 - When he/she clicks a square, if it is empty, it will print "S" on it if the checkbox is checked; otherwise it will print an "O".
 - After the user clicks, the computer takes the charge and prints an "S" or "O" randomly on a random empty square.
 - The squares that the user or computer played in one turn should be shown in thick line.
- The checkbox/buttons/spinners have the following functions:
 - First Activity:
 - * When the user clicks "S", there is no change but the click behavior of the user changes.
 - * When the user clicks "Restart Game", the first activity is reloaded and the game restarts with the new options.
 - * When the user clicks "Show Options", the second activity is started.
 - * When the user clicks "Load Game", the board will be loaded from the "game.txt" file which is stored in the raw folder.
 - Second Activity:
 - * When the user changes the value of the spinner, the value of the spinner must be displayed on the corresponding Editbox.
 - * When the user clicks "Restart Game", the first activity is reloaded and the game restarts with the new options.
 - * When the user clicks "Go back", there is no change but only the program returns to the first activity.
- You should use tag attribute while defining the behavior of the spinners. You can not write two separate functions
- You should declare all strings in the program in the strings.xml.
- You should save width and height in the SharedPreferences and load them.
- Two examples of the "game.txt" is given below. The width and height are equal, but can be changed. Empty space is represented with "_".
First one:


```
S_O_
__S_
O_S_
_O__
```

 Second one:


```
___O_
____
__S__
___O_
_____
_____
```